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# Homology Modelling and Active-Site Engineering of *Burkholderia cepacia* Biphenyl-2,3-Dioxygenase for Enhanced PCB Degradation

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## Abstract

A **homology model** of the *Burkholderia cepacia* **biphenyl-2,3-dioxygenase (BPDO)**  $\alpha$ -subunit (**BphA**; UniProt Q46372) was constructed using **SWISS-MODEL**, with the *Paraburkholderia xenovorans* LB400 P4 variant (**PDB: 2XSH**, Chain A) as the structural template. Three **PCB congeners**—**2,2'-dichlorobiphenyl (22DCB)**, **3,3'-dichlorobiphenyl (33DCB)**, and **4,4'-dichlorobiphenyl (44DCB)**—were manually positioned in the active site using **PyMOL**; the closest **Fe–ligand distances** were measured as 2.352 Å, 3.331 Å, and 2.637 Å, respectively. Interaction analysis identified a conserved **hydrophobic cluster** (PHE227, PHE376, PHE382, TRP390) and the **catalytic iron-coordinating triad** (HIS233, HIS239, ASP386) as the principal active-site determinants. A targeted **PHE227→ALA** substitution, performed *in silico* using the PyMOL Mutagenesis Wizard, reduced the 33DCB Fe–ligand distance from 3.331 Å to approximately 2.7 Å, demonstrating improved **catalytic alignment**.

## 1. Introduction

**Polychlorinated biphenyls (PCBs)** are persistent priority pollutants whose aerobic microbial catabolism is initiated by **biphenyl-2,3-dioxygenase (BPDO)**, a multicomponent **Rieske non-heme iron oxygenase** that catalyses *cis*-dihydroxylation of the biphenyl ring [?, 1–3]. The large  $\alpha$ -subunit (**BphA**) governs **substrate specificity**; a small number of **active-site exchanges** are sufficient to shift congener preference between **ortho-**, **meta-**, and **para-substituted PCBs** [2, 4]. Because no crystal structure exists for *B. cepacia* BphA, I constructed a **homology model**, manually docked three representative congeners, and performed targeted *in silico* mutagenesis to improve binding of the sterically demanding meta-substituted **33DCB**.

## 2. Materials and Methods

### 2.1 Template Selection and Homology Modelling

The target sequence (**Q46372**) was submitted to **SWISS-MODEL. BLAST** against the PDB identified the *P. xenovorans* LB400 P4 variant (**PDB: 2XSH**, Chain A) as the optimal template on the basis of **sequence identity**, coverage, and biological relevance (sequence similarity over 70% and identity over 30%).

### 2.2 Ligand Docking and Interaction Analysis

3D structures of **22DCB**, **33DCB**, and **44DCB** were loaded into **PyMOL** alongside the BphA model. Each ligand was manually positioned near the **catalytic Fe(II) centre**, optimising **steric compatibility** and **geometric alignment**. Using PyMOL with Python scripting, **Fe–ligand distances**, residues within 4 Å, and **steric clashes** were extracted and exported as CSV.

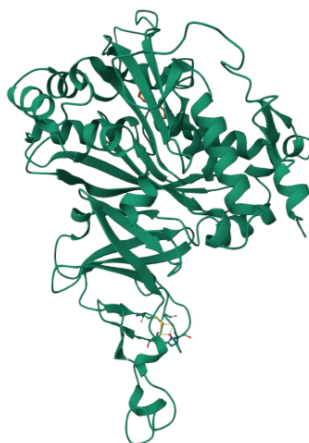
### 2.3 In Silico Mutagenesis

Based on clash analysis identifying **PHE227** as the primary **steric constraint** for 33DCB, a **PHE227→ALA** substitution was executed via the **PyMOL Mutagenesis Wizard**. The ligand was repositioned and the Fe–ligand distance re-measured.

## 3. Results and Discussion

### 3.1 Homology Model

The SWISS-MODEL run produced a full-chain BphA model with the canonical  $\alpha\beta$ -barrel fold of Rieske oxygenase  $\alpha$ -subunits, incorporating the **Rieske iron-sulfur cluster domain** and the **catalytic mononuclear iron site**.

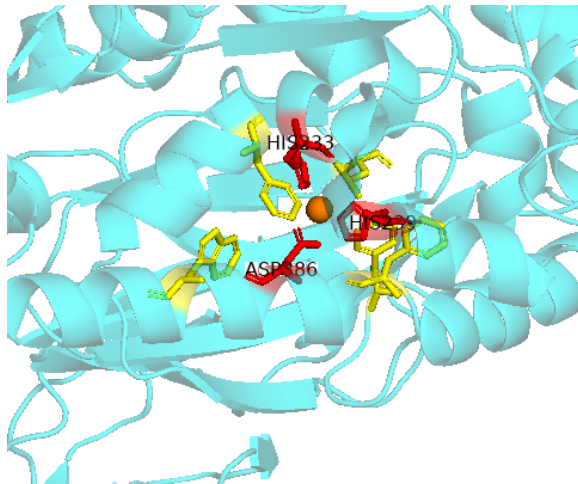


(a) BphA homology model (SWISS-MODEL).

**Figure 1:** Three-dimensional ribbon representation of the *B. cepacia* BphA  $\alpha$ -subunit generated by SWISS-MODEL.

### 3.2 Active-Site Architecture

The **hydrophobic substrate-binding pocket** is lined by an **aromatic cluster**—**PHE227**, **PHE376**, **PHE382**, **TRP390**—and the canonical **2-His-1-carboxylate facial triad** (**HIS233**, **HIS239**, **ASP386**) coordinating the catalytic iron (Figure 2). These residues were consistently identified within 4 Å across all three docking experiments.



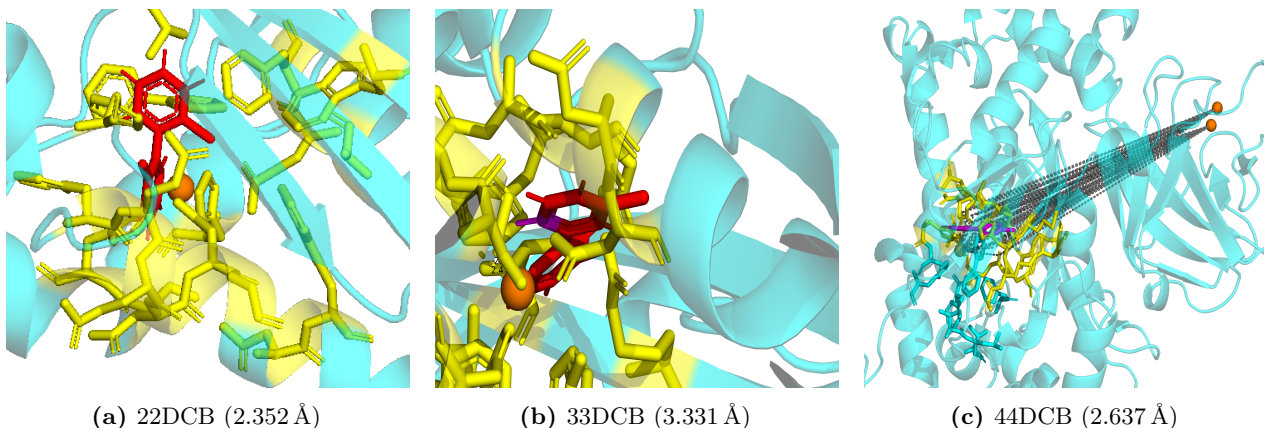
**Figure 2:** Active-site architecture of *B. cepacia* BphA showing the **catalytic Fe** (sphere), **2-His-1-carboxylate coordination triad** (HIS233, HIS239, ASP386), and **hydrophobic pocket-lining residues** (PHE227, PHE376, PHE382, TRP390).

### 3.3 Comparative Docking of PCB Congeners

**Fe–ligand distances** and accommodation quality are summarised in Table 1. **22DCB** achieved the shortest distance (2.352 Å), indicating **favourable catalytic alignment** despite ortho steric bulk. **33DCB** gave the largest distance (3.331 Å); **PHE227** and **ASP386** were the principal **steric constraints**. **44DCB** showed intermediate binding (2.637 Å), stabilised by **hydrophobic contacts** with PHE376, PHE382, and TRP390.

**Table 1:** Fe–ligand distances and accommodation quality for the three PCB congeners.

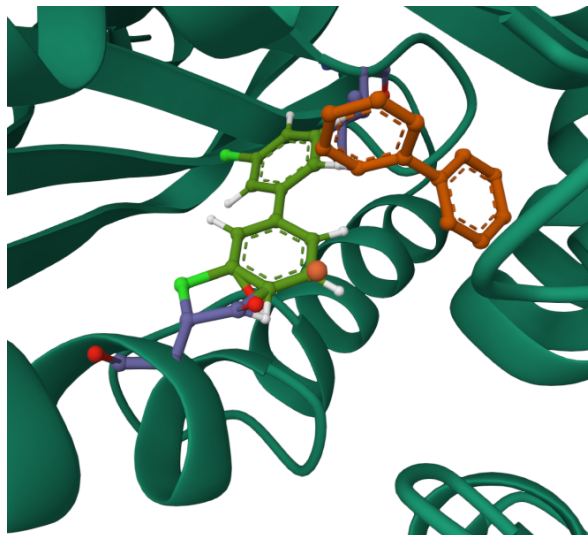
| Congener | Substitution Pattern | Fe–Ligand Distance | Accommodation |
|----------|----------------------|--------------------|---------------|
| 22DCB    | <i>ortho</i>         | 2.352 Å            | Favourable    |
| 33DCB    | <i>meta</i>          | 3.331 Å            | Restricted    |
| 44DCB    | <i>para</i>          | 2.637 Å            | Moderate      |



**Figure 3:** Comparative **active-site accommodation** of the three dichlorobiphenyl congeners within the BphA catalytic pocket. Fe–ligand distances are indicated in each panel.

### 3.4 *In Silico* Mutagenesis: PHE227→ALA

**PHE227** imposed the dominant **steric constraint** on 33DCB. Substitution to **Alanine**—the smallest chiral amino acid—was performed using the **PyMOL Mutagenesis Wizard**. After repositioning 33DCB in the enlarged pocket, the **Fe–ligand distance** decreased from 3.331 Å to  $\sim 2.7$  Å, with a more **planar biphenyl ring orientation** expected to favour productive **oxygenation**. This is consistent with literature showing that few amino acid exchanges suffice to substantially shift **BPDO congener preference** [2, 4].



**Figure 4:** **PHE227→ALA** mutagenesis improves 33DCB accommodation. The substitution enlarges the substrate-binding cavity, reducing the Fe–ligand distance from 3.331 Å to  $\sim 2.7$  Å.

## 4. Conclusions

A validated **homology model** of *B. cepacia* **BphA** was used to characterise the differential **active-site accommodation** of three **PCB congeners**. **22DCB** bound most favourably (Fe–ligand: 2.352 Å); **33DCB** was **sterically restricted** (3.331 Å); **44DCB** showed intermediate binding (2.637 Å). A single **PHE227→ALA** substitution expanded the pocket and improved 33DCB **catalytic proximity** to  $\sim 2.7$  Å, providing a **structural rationale** for targeted engineering.

## References

- [1] K. N. Timmis *et al.*, *Annu. Rev. Microbiol.*, 48:525–557, 1994.
- [2] K. Furukawa *et al.*, *J. Bacteriol.*, 186(16):5189–5196, 2004.
- [3] M. Sylvestre, *Int. Biodeterior. Biodegrad.*, 54:153–162, 2004.
- [4] P. Kumar *et al.*, *J. Mol. Biol.*, 405(2):531–547, 2011.